

SJAA Ephemeris Bulletin

✱ June 2025

Celebrate Summer

There are many reasons to celebrate in the month of June, not least of which is the summer solstice for the Northern Hemisphere. This is the time (June 20th at 7:42pm this year) when the North Pole is most tilted towards the Sun. This day will contain the longest stretch of daylight of the year: about 15 hours and 48 minutes at SJAA's home-base of Houge Park.

But, the solstice isn't the only thing to celebrate in the summer. Many astro-friendly folks celebrate birthdays this month, including Charles Messier of Messier Catalog fame, Nobel laureate Kip Thorne of LIGO fame, and astronomer/comet-hunter Carolyn Shoemaker. As if that's not enough, SJAA's calendar is chock full of observing opportunities, astro 101 talks, and more. We've also got many hours of great content on YouTube, to fill the time waiting for astronomical dark on these long summer days.

Keep looking up!

Carl Svensson
Vice-President
Ephemeris Editor

Target of the Month - M51

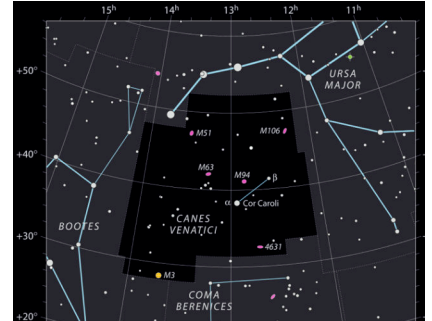
by Joe Fragola

Also known as the "Whirlpool Galaxy," M51 was discovered by Charles Messier on October 13, 1773. A companion galaxy, NGC 5195, was discovered in March 1781 by Pierre Mechain. NGC 5195 is a dwarf galaxy that is interacting with the Whirlpool Galaxy. Originally classified as a "spiral nebula," M51 and other "spiral nebulae" were not recognized as galaxies until Edwin Hubble found evidence that they were so far away that they must be entirely separate galaxies.

In September 2020 the Chandra X-Ray Observatory detected a candidate exoplanet located within M51. Confirmed in October 2021, this is the first known instance of a planet outside our Milky Way galaxy.

You can find the Whirlpool Galaxy in the constellation Canes Venatici, the Hunting Dogs. At magnitude 8.4, this galaxy can be seen with binoculars from dark skies. Look for a fuzzy object with an apparent size of 13' x 11', 3.5° southwest of Alkaid, the last star in the "handle" of the Big Dipper.

Magnitude	8.4
Size	77 kly
Object Type	Spiral Galaxy
Constellation	Canes Venatici
Distance	29-31 Mly
Age	1 - 2 billion years



Top: Canes Venatici sky chart from theskylive.com. Bottom: M51 by Joe Fragola on 5.17.2025 from RCDO. 44 minutes of exposures on Celestron Origin.

Volunteer Opportunities

There are many volunteer opportunities in SJAA! Please do not hesitate to reach out to the volunteer coordinator (volunteerchair@sjaa.net) to inquire about opportunities. Below are a few examples of programs that could use help:

Imaging SIG The SJAA Imaging SIG (Special Interest Group) welcomes members to give talks in our monthly imaging meetings on topics related to astro-imaging. World-class expertise not required! Example topics include: comet astrophotography, solar imaging, PixInsight skills, NINA tutorial, remote imaging, collimating telescopes, "my favorite astro gadgets", the science behind astrophotographs, "I love my mount/scope/software/...". Please don't be modest, we need you! For past talks see our YouTube channel (@sj-astronomy7497). Please email imagingSIG@sjaa.net if you're interested.

Loaner Program Our loaner program is looking for one to two more volunteers to help members check out gear and to demonstrate the set-up and usage of loaned items. The only requirement is an interest in astro gear and availability on one Sunday per month. Email loaners@sjaa.net if you are interested.

SSP The School Star Party Program is looking for more regular volunteers. We provide fun and exciting observing opportunities for schools. The ideal volunteer should have a telescope, some basic astronomy knowledge, and enjoy inspiring people. Interested parties should be available for the events and can contact ssp@sjaa.net for more information.

Get it Fixed!

Are your telescope's images not what they use to be? Do stars look like small comets? The likely cause is that the optics are no longer collimated and the images you are observing are off axis. This can occur if a telescope has been disassembled for cleaning or just due to vibration during transportation. SJAA has a monthly telescope "fix it" clinic for its members free of charge as one of the many member benefits. All you need to do is sign up via our website to make an appointment. Go to <https://www.sjaa.net/programs/fixit/> to sign up for an appointment.



SJAA June 2025

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1 Solar, Loaner, Fix-It 2:00PM Kip Thorne's Birthday (1940) Mars 1.4° S of Moon  39%	2  50%	3 First Quarter Moon  60%	4  69%	5  78%	6 Astro 101 Talk (Online) 7:00PM Spica 0.5° N of the Moon  85%	7 SJAA Board Meeting 6:00PM  91%
8 Mercury 2° N of Jupiter  96%	9  96%	10 Antares 0.3 N of Moon  99%	11 Full Moon  100%	12  99%	13  96%	14 Binocular Stargazing 9:30PM Pluto 0.1° N of Moon Flag Day  90%
15  83%	16 Mars 0.8° N of Regulus  74%	17 Imaging SIG (Online) 7:30PM  64%	18 Neptune 2° S of Moon Saturn 3° S of Moon Last Quarter Moon  53%	19  41%	20 Summer Solstice 6:42PM Astro 101 Talk 8:00PM In Town Star Party 9:30PM  30%	21 Starry Nights Star Party 9:30PM  19%
22  11%	23 Moon 0.6° N of the Pleiades  4%	24 Carolyn Shoemaker's Birthday (1929)  1%	25 New Moon  0%	26 Mercury 3° S of Moon Lyman Spitzer's Birthday (1914) Charles Messier's Birthday (1730)  3%	27  8%	28 Mendoza Ranch Observing 9:30PM  15%
29 Saturn 1° S of Neptune Regulus 1.5° S of Moon Mars 0.2° S of Moon  24%	30 George Ellery Hale's Birthday (1868)  33%					

Times and dates are subject to change. Please check
sjaa.net and www.meetup.com/sj-astronomy for the latest
 updates.

Meteor Shower

Dark Sky Window